BrightCart Cloud

E-Commerce Retention Rescue Strategy Recommendation Report

Strategy Recommendation Report

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Dataset: UCI Online Retail (Dec 2010 – Dec 2011)

Tools : MySQL · Python · Power BI

GIT HUB: [https://github.com/pasinduheshan44/Bright-Retention-Analysis]

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1. Executive Summary

The BrightCart Cloud is going through the silent crisis of retention. New customer acquisitions are remaining constant, but there is a drop in the repeat customer purchases. The total revenue may look flat, due to seasonal peak months like November 2011, which saw a revenue peak of £1,162,000, but actually the issue lies deeper. The firm keeps losing its repeat customers all the time and is constantly acquiring new customers.

The month-on-month analysis of the performance of 4,338 customers for 13 months indicates that the Month-1 retention has decreased from 36.6% for the December 2010 cohort to just 11.2% for the November 2011 cohort, i.e., a drop of about 69%. The analysis by RFM technique shows that there are 114 Lost -High value customers accounting for £223,575 of at-risk revenue and another 608 "About to sleep" customers representing £336,037 of revenue. Hence, there is an amount of £559,612 of at-risk revenue, which can easily be recovered.

There are three actions to take based on the analysis done above:

(1) Automated on-boarding email sequence to recover Month-1 retention,

(2) Allocation of marketing budgets in favour of high retention products and countries,

(3) Win-back campaign to retain those high-value lost customers.

2. Business Question

"Why are repeat purchases declining among previously high-value customers, and which products, countries, and customer segments should the company prioritize to improve long-term revenue retention?"

Such an analysis topic was selected to address the main symptoms mentioned in the scenario about BrightCart Cloud. These include declining repeat purchases, constant sales despite discounts, and lack of information about the place from which the churn happens. The reason behind this is that by focusing only on loyal customers who used to make purchases before, i would be targeting the most effective group.

The problem statement itself should produce three outputs. First of all, the analysis aims to determine why repeat purchases decline. In addition, the analysis aims prioritise the segments based on the type of product and country. Finally, I must formulate what the company needs to do to solve this problem the following month. Unlike any general question about customer retention, this analysis is financially viable.

3. Data Preparation & Cleaning

3.1 Data Description

The data used in the analysis is UCI Online Retail data set which represents a transaction record from a UK-based online retailer between 1 December 2010 and 9 December 2011. This raw data set includes 541,909 records and has 8 variables namely, InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. Data is loaded into MySQL database using Python (mysql-connector-python) in segments of 5000 rows after that it was analyzed in XAMPP MySQL using 11 structure analysis tables and finally exported to Python for visualization.

3.2 Missing Values Analysis

Before the cleaning process, two variables contained missing data. Customer ID has 135,080 missing values (24.9%) and Description contains 1,454 missing data points (0.3%). All other variables, InvoiceNo, StockCode, Quantity, InvoiceDate, UnitPrice and Country have no missing data points.

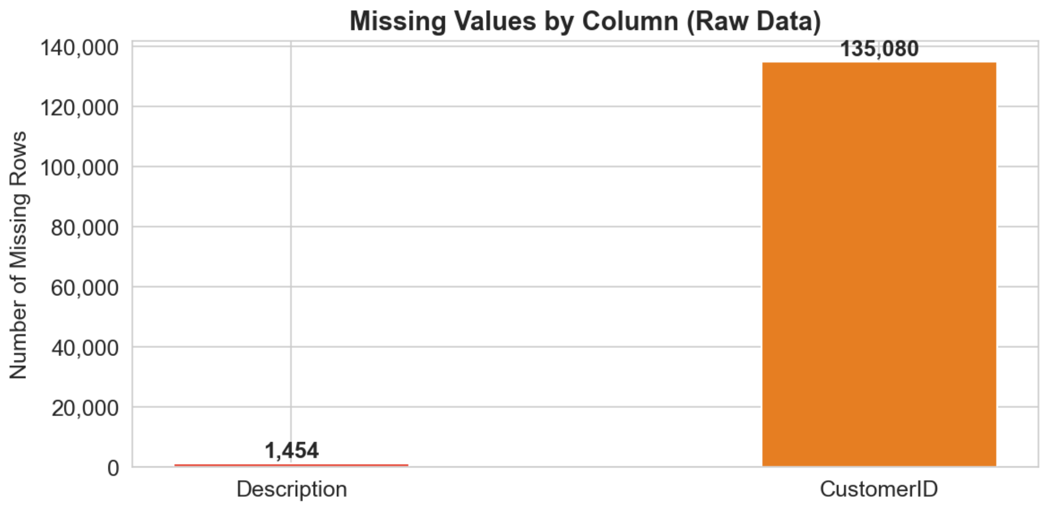

Figure 1 Missing Values by Columns

3.3 Application of Cleaning Rules

There were five cleaning rules to ensure that there is no discrepancy in the analysis. Justification is provided below:

Table 1 Cleaning evidence

Stage	Count	Removed	Reason(rule)
Raw Data	541,909	-	Baseline - Online_Retail.csv
Removed: Missing CustomerID	135,080	135,080	Cannot Tracking without ID
Removed: Missing Description	1,454	1,454	Product name is required
Removed: Cancelled Invoices (C %)	9,288	9,288	Not real purchases
Removed: Negative / Zero Quantity	~1,800	~1,800	Not real sales
Removed: Zero / Negative Price	~52	~52	Internal adjustments
TOTAL CLEAN ROWS	397,884	144,025 Removed	Analysis used in all

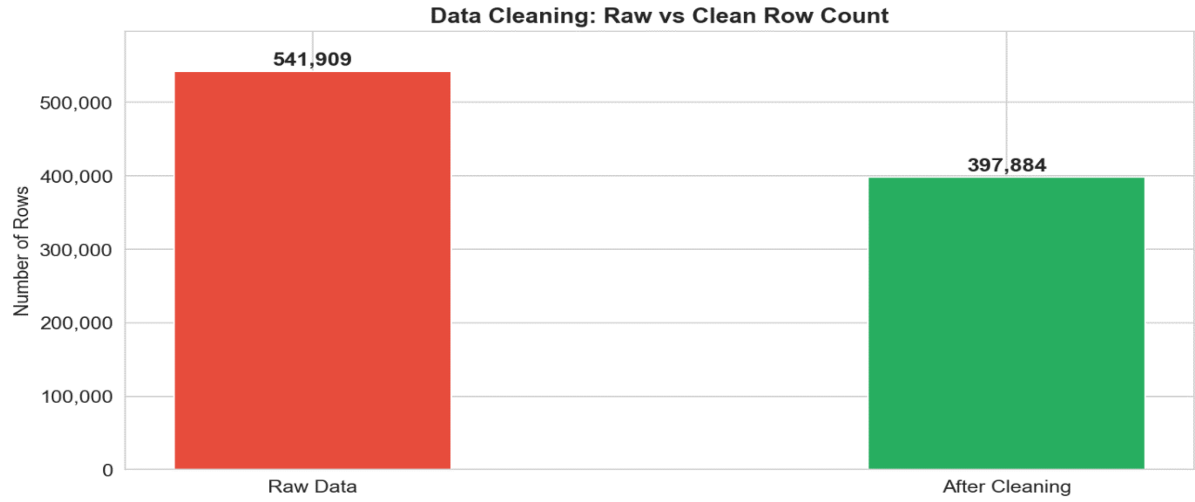

Figure 2 Data cleaning summary

3.4 Data Pipeline

The cleaned dataset was organized in eleven MySQL tables consisting of orders_clean, customer_first_purchase, monthly_customer_revenue, cohort_size, cohort_retention, cohort_heatmap, product_summary, country_summary, rfm_scores, and customer_segments. Visualizations were done using Python (NumPy, pandas, matplotlib, and seaborn)

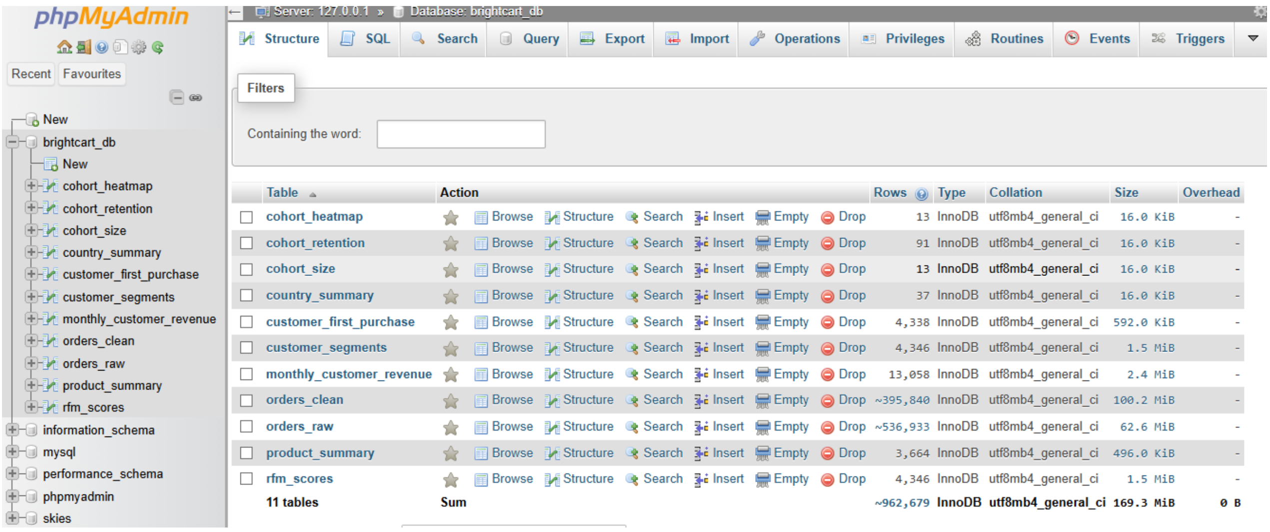

Figure 3 My Sql tables creation evidence

4. Analysis of Retaining Customer Base

4.1 How to Conduct Cohorts Analysis

Cohorts Analysis is an analysis technique intended to organize customers into groups based upon their first month of being a customer, then track how many of these customers returned for every month thereafter. M0 is 100% (starting cohort) and M1 is the percentage of cohorts that returned for the next month after the first purchase (M1). A decline in the M1 rate among cohorts indicates that retention will decline regardless of how many new customers are being acquired. The formula used to calculate retention is:

Retention% = Customers Active in Cohort Month (N)/Total Number of Customers in Cohort Month (0) * 100

4.2 Result of Heat map for Cohorts Analysis

The cohort's retention percentages for December 2010 and December 2011 are displayed in the sample heat map below. Each cell is color coded based on retention percentages, green cell indicates percentage greater than 30%, yellow cells indicate percentages between 20%-30% and red cells indicate percentages less than 20%.

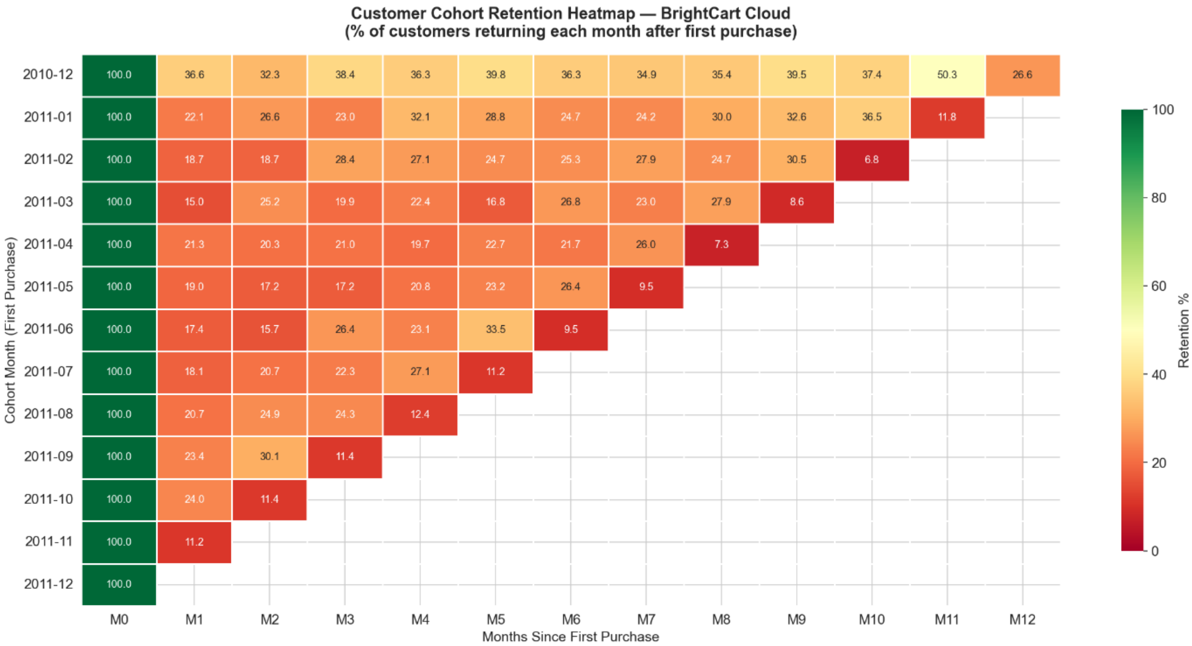

Figure 4 Customer Cohort Retention Heatmap

Every row shows a cohort of customers based on first-time purchases.

All customers that purchased for the first time in the month and cohort belong to M0 (100%).

All customers that purchased again in later months belong to M1, M2, M3, and so on. A lower percentage indicates that customers are weakly retained and do not consistently make repeat purchases.

4.3 Key Findings

- Which cohort saw the fastest decline?

The cohort of 2011-11 (Month 1) has the lowest rate of customer retention at 11.2%.

The subsequent cohorts (2011-10 and 2011-09) have a lower rate of customer retention than earlier cohorts, indicating that there may be an overall quality decline in these customers, a lack of quality in the on boarding process, or minimal post-purchase engagement.

- In which month did the drop-off occur?

The drop will be highest between M0 and M1 for nearly all cohorts.

For instance, from 2011-03 (Month 0) to 2011-0 (Month 1) will see a decline from 100% down to 15% of customers retaining. The majority of customer loss occurs immediately following the initial purchase.

- Does the revenue retention story look similar to customer retention?

Absolutely.

Cohorts with lower customer retention generally provide lower amounts of repeat revenue over time. Therefore, lower numbers of repeat customers acquired will eventually result in less long-term revenue retention as measured as aged dirty customers who have previously purchased.

Business Insights

According to the Heatmap, BrightCart Cloud faces its key challenge in maintaining customer loyalty rather than attracting new customers. A significant number of customers from all cohort groups tend to drop off shortly after their first purchase; newer cohorts tend to show lower retention than older cohorts. Focus on enhancing first-time customer experience, improving Onboarding processes and re-engaging with at-risk customers will likely lead to higher rates of repeat purchases by customers over time and ultimately result in greater long-term revenue retention

CohortMonth	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
2010-12	100.0%	36.6%	32.3%	38.4%	36.3%	39.8%	36.3%	34.9%	35.4%	39.5%	37.4%	50.3%	26.6%
2011-01	100.0%	22.1%	26.6%	23.0%	32.1%	28.8%	24.7%	24.2%	30.0%	32.6%	36.5%	11.8%	-
2011-02	100.0%	18.7%	18.7%	28.4%	27.1%	24.7%	25.3%	27.9%	24.7%	30.5%	6.8%	-	-
2011-03	100.0%	15.0%	25.2%	19.9%	22.4%	16.8%	26.8%	23.0%	27.9%	8.6%	-	-	-
2011-04	100.0%	21.3%	20.3%	21.0%	19.7%	22.7%	21.7%	26.0%	7.3%	-	-	-	-
2011-05	100.0%	19.0%	17.2%	17.2%	20.8%	23.2%	26.4%	9.5%	-	-	-	-	-
2011-06	100.0%	17.4%	15.7%	26.4%	23.1%	33.5%	9.5%	-	-	-	-	-	-
2011-07	100.0%	18.1%	20.7%	22.3%	27.1%	11.2%	-	-	-	-	-	-	-
2011-08	100.0%	20.7%	24.9%	24.3%	12.4%	-	-	-	-	-	-	-	-
2011-09	100.0%	23.4%	30.1%	11.4%	-	-	-	-	-	-	-	-	-
2011-10	100.0%	24.0%	11.4%	-	-	-	-	-	-	-	-	-	-
2011-11	100.0%	11.2%	-	-	-	-	-	-	-	-	-	-	-
2011-12	100.0%	-	-	-	-	-	-	-	-	-	-	-	-

5. Revenue Trend Analysis

5.1 Monthly revenue Overview

During the 13-month analysis was approximately £8.9 million GBP. The total monthly revenue ranged between a low of £447,000 for February 2011 and a High of £1,162,000 for November 2011 during pre-Christmas shopping behaviour. Monthly average order values ranged from £18/£30, with the December 2011 product being sold outside of this range (£29.96) due to the sale of high dollar items for Christmas gift giving.

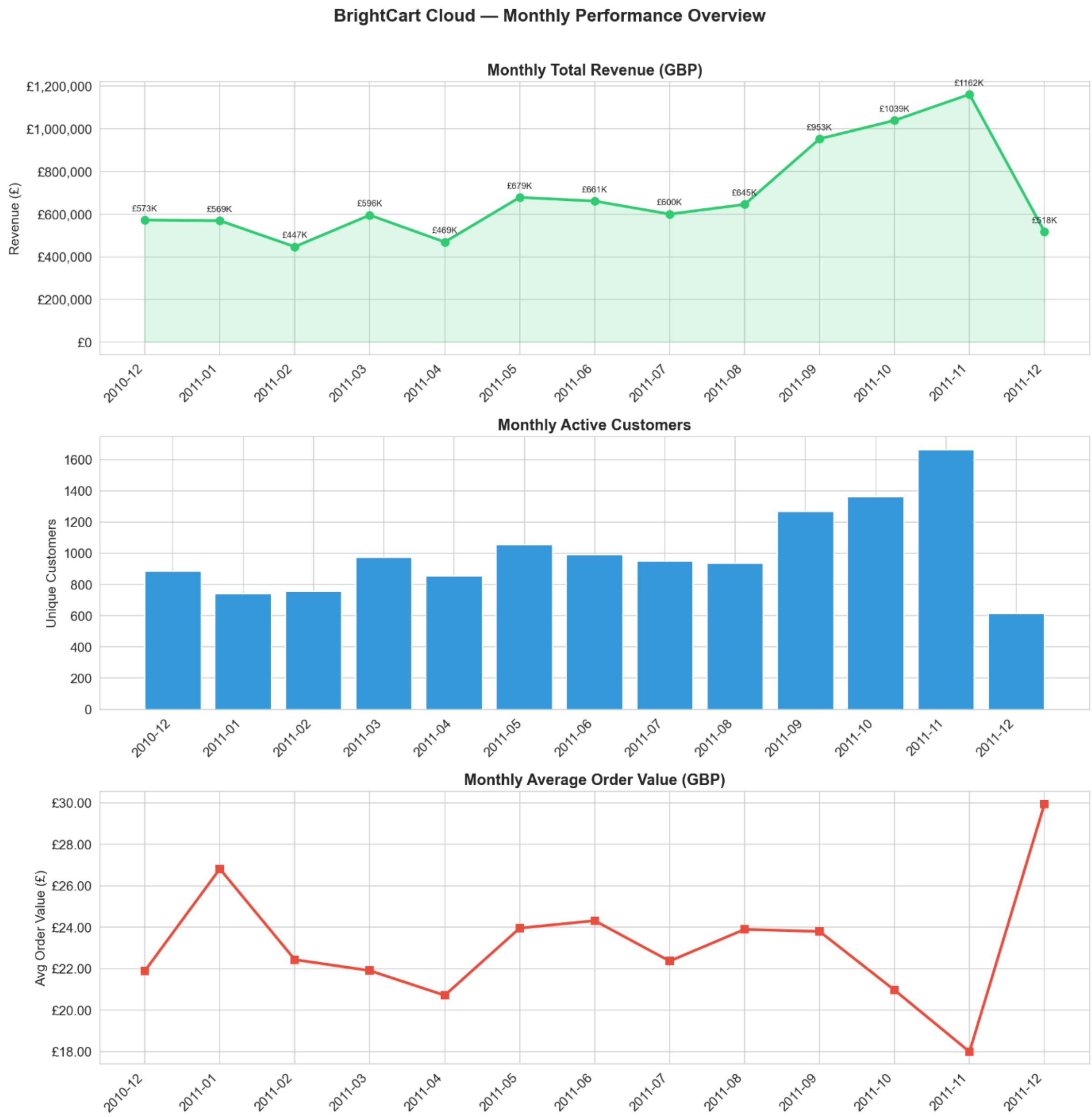

Figure 5 Monthly Performance Overview: Total Revenue (GBP), Active Customers, and Average Order Value

5.2 New vs Repeat Customer Revenue

By separating revenue by customer type I am able to illustrate the underlying structure of the business that is hidden by aggregate numbers. In December 2010 (the first month), there were no existing repeat buyers (civil customers). By December 2011 repeat customers made up 93% of the revenue sold, demonstrating that the vast majority of monthly revenue was derived from repeat customers.

The trend of repeat purchase rates does not demonstrate consistent improvement month to month. The rate increased from a low of 43.7% in January 2011 to a High of 81.9 % in August 2011, declining to 73.8% in October 2011 before increasing to January 2012. The decrease reflects the month of the highest amount of new customers being acquired (September through November 2011) which suggests to me that buyers who were purchased during this time frame were not retained very well.

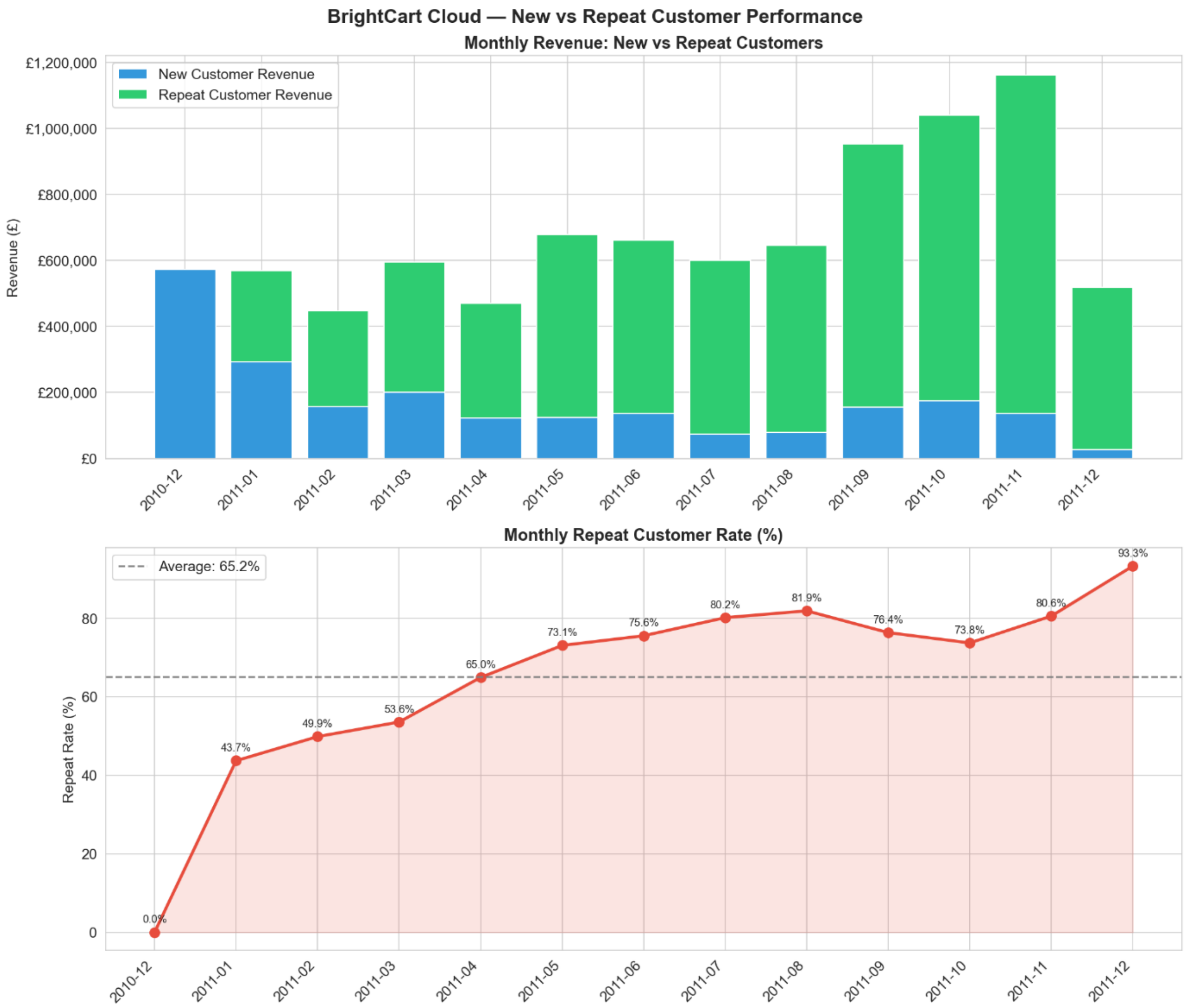

Figure 6 New vs Repeat Customer Revenue by Month and Monthly Repeat Purchase Rate (%)

5.3 Revenue Insight

The increase in the percentage of repeat buyers to 93.3% for December 2011 sounds like a positive trend, but it is largely just a product of the way numbers work. In December 2011, most of the active customers are long-standing repeat customers, as there will not be another time frame to see if new cohorts will also repeat purchases, therefore making the repeat purchase rate look artificially high. To get a more precise view of what is happening with new cohorts, you can refer to the cohort analysis in Section 4, which shows that new cohorts are losing customers at an increasing rate (from their first month of activity) and, therefore, if no action is taken, the company will experience a weakening revenue pipeline in 2012.

6. Customer Segment Analysis (RFM)

6.1 RFM Methodology

RFM analysis of 4,338 customers scores across 3 dimensions using quartile scoring (score of 1-4, with a score of 4 being the best) and includes:

(i) Recency (R) – Days from last order to 10 December 2011 (lower date = higher score);

(ii) Frequency (F) – Number of unique orders placed (higher number = higher score);

(iii) Monetary (M) – Pound/GBP amount of all orders (higher amount = higher score).

The combined score (R+F+M) for each customer is from 3 (worst) to 12 (best). Based upon their combined scores, customers have been segmented into 7 segments relevant to the business.

6.2 Segment Results

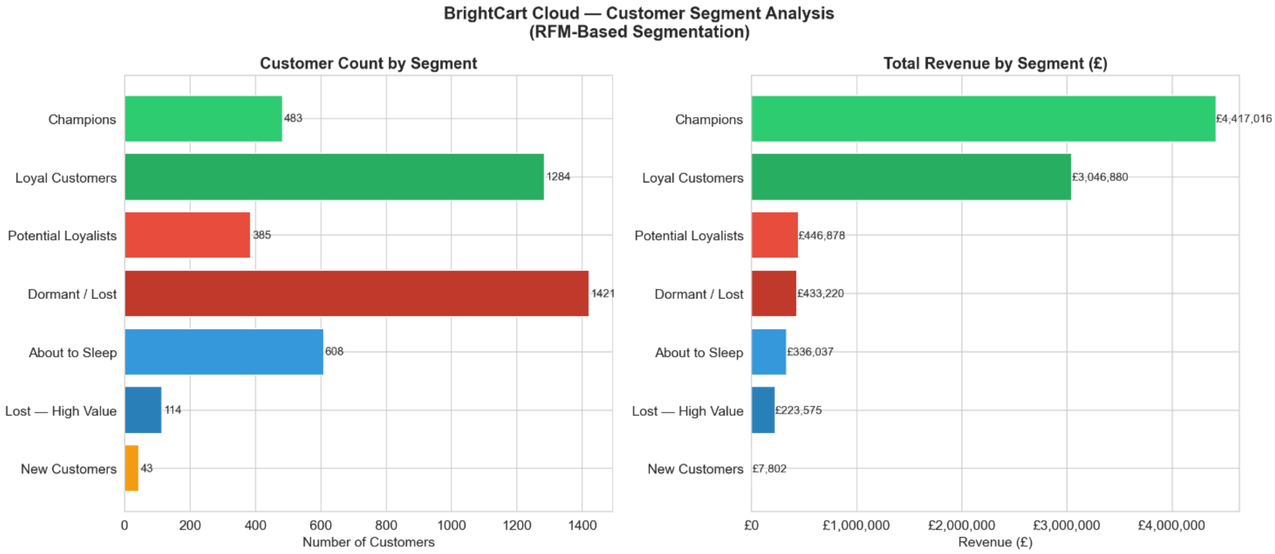

Figure 7 Customer Segment Analysis (RFM)

Table 2 Customer Segment Distribution

Segment	Count	% Cust	Revenue	% Rev	Recommended Action
Champions	483	11.1%	£4,417,016	49.6%	Reward & upsell - highest priority
Loyal Customers	1,284	29.6%	£3,046,880	34.2%	Loyalty programme - keep engaged
Dormant / Lost	1,421	32.8%	£433,220	4.9%	Low priority - survey or remove
About to Sleep	608	14.0%	£336,037	3.8%	Re-engagement campaign
Potential Loyalists	385	8.9%	£446,878	5.0%	Nurture to 2nd purchase
Lost - High Value	114	2.6%	£223,575	2.5%	Win-back - strong incentive
New Customers	43	1.0%	£7,802	0.1%	Onboarding email sequence

6.3 Key Segment Findings

- Champions (11.1% of customers) are responsible for generating £4,417,016-which accounts for 49.6%-of total revenue; this segment should be the highest priority for BrightCart Cloud to protect.
- Loyal customers (29.6%) have generated an additional £3,046,880 (34.2%), combining for 83.8% of all revenue from Champions + Loyal Customers.
- High-value Lost customers (114/2.6%) have £223,575 in historical revenue lost-and were previously confirmed high-spending customers-make these the priority target market for re-engaging.

- "About to sleep" (608 customers) are now at immediate risk of lapsing within the next 30-60 days without intervention, with a value of £336,037.
- Dormant/Lost customers make up 1,421/32.8% and only create £433,220-4.9%-in revenue, and represent the cost to BrightCart Cloud of not successfully retaining customers over time.

6.4 RFM Scatter Analysis

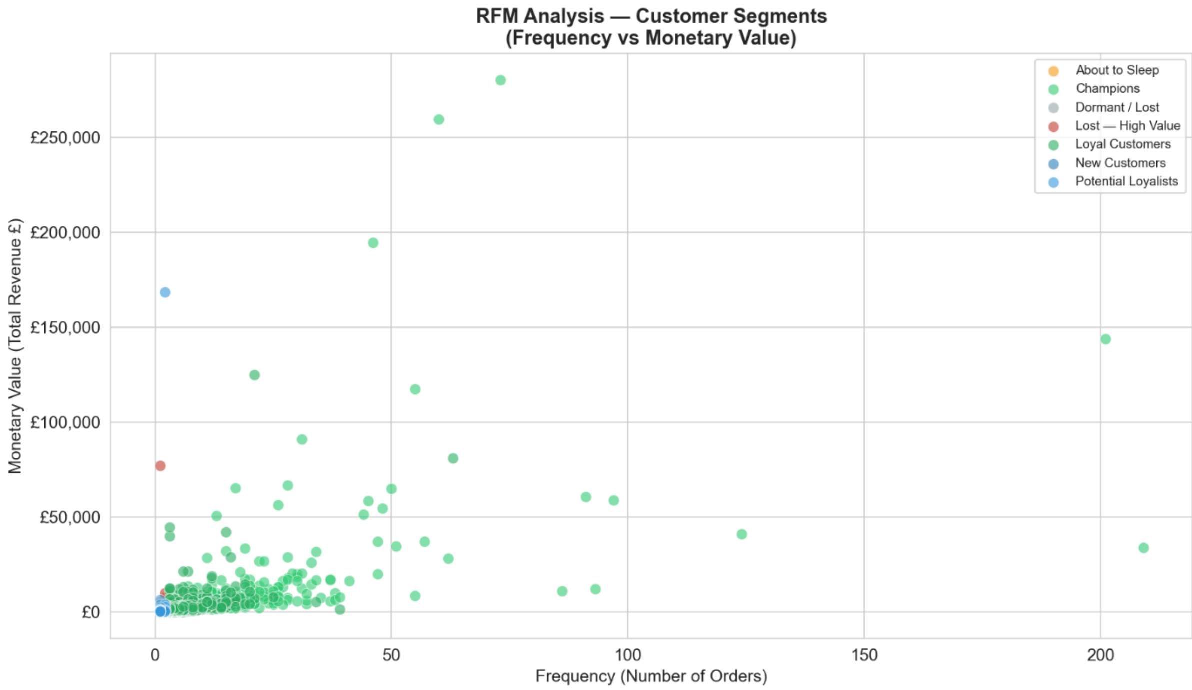

Figure 8 RFM Scatter plot

Key Findings

- Who is most valuable for the business?

Champions and Loyal Customers customer segments bring the most money to the business and make the largest number of transactions.

They form the most valuable customer segments that are essential for maintaining revenue.

- Which customers should be kept?

Lost – High Value segment includes customers that used to bring a lot of money but currently demonstrate low purchasing frequency.

Their loss may negatively influence repeat revenue.

- What conclusion does this help us draw about repeat purchases declining?

The scatter plot clearly indicates that revenue comes from a rather small number of frequent customers.

Any decrease in their activity may cause a dramatic drop in repeat revenue.

It proves the idea that retaining valuable customers matters more than attracting new ones.

Business Insight

According to the RFM analysis, BrightCart Cloud relies a lot on the Champions and Loyal Customers for their sustainable revenue in the future. On the other hand, some high-value customers have been lost and are currently included in the Lost - High Value category, which presents an issue of customer retention. In order to ensure future revenue retention, the business needs to retain the high-value customers, revive the lost customers, and engage the Potential Loyalists.

7. Product & Country Analysis

7.1 Product Analysis

The analysis of products at the product level demonstrates the various factors related to products that create the highest revenue and the highest number of repeat customer purchases for products. Products with both very high revenue and high repeat purchasing customers are the strongest retention anchors for customers returning to make additional purchases.

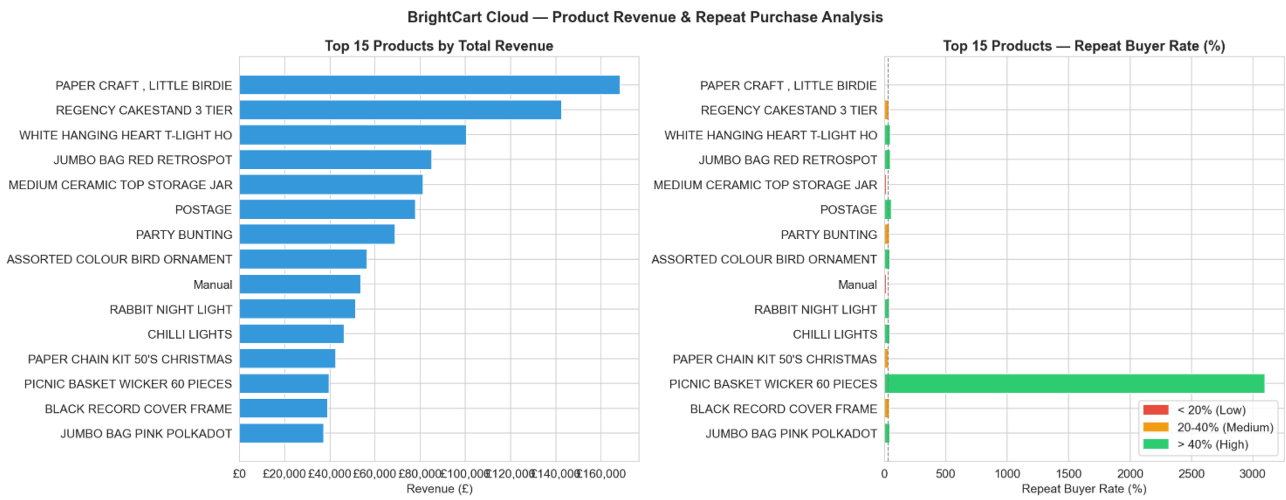

Figure 9 Top 15 Products by Total Revenue and Repeat Buyer Rate %

Key findings from the product review include the following:

- Overall highest revenue product is the PAPER CRAFT, LITTLE BIRDIE with approximately £163,000 in revenue, more than double the revenue of the 2nd highest revenue product

- The 2nd and 3rd highest revenue products are the REGENCY CAKESTAND 3 TIER and the WHITE HANGING HEART T-LIGHT HOLDER, respectively
- The highest repeat purchase rate of the top 15 products is for the PICNIC BASKET WICKER 60 PIECES, which indicates that there is a strong customer base for this product
- Most of the other high revenue products have a repeat purchase rate of under 20% (red) and as such do not provide to the customer any significant retention due to customers only making a single purchase and then not returning to make a repeat purchase.

Thus the cohort analysis supports the previously identified issue of an Onboarding gap for customers by these types of products.

7.2 Country Analysis

The United Kingdom is the market leader, contributing more than 80% of total revenues. However, while some international markets have fewer overall customers than the UK, they tend to have higher repeat rates after an initial purchase.

Key findings by country:

• United Kingdom - highest revenue generation (by volume) but also has the most churned customers due to market size

• Netherlands, Germany, France & Eire are the top five priority international markets, based on a combination of revenue (by volume) & repeat customer rates

• International markets are an under-invested growth opportunity (i.e., customers in these markets will generally buy more frequently than the average British customer).

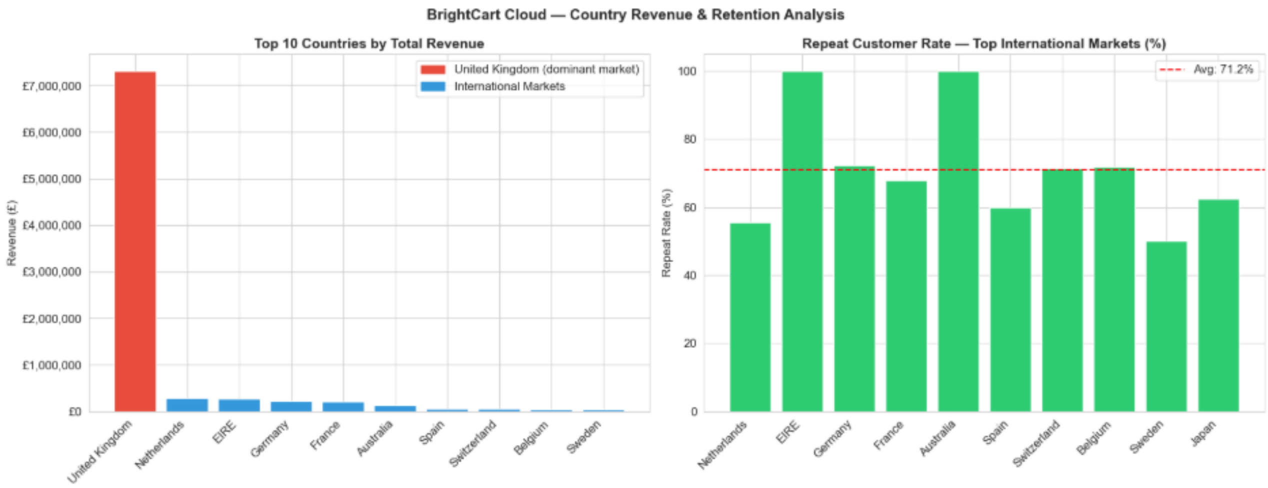

Figure 10 Country Revenue Analysis and Repeat Customer Rate by Top International Markets

8. Recommendations for Strategic Action

After compiling all of the data, there are three recommended strategic actions to take. All outlined recommendations can be implemented within 30-days and assigned to a designated team with defined benchmarks for success.

Recommendation 1: to stop the decline of the Onboarding Stage - recovering Month-1 Retention.

Description of Evidence

The cohort retention Heatmap in (Figure 4) for evidence of all cohorts experiencing the largest decline, in terms of retention rate between Month 0 and Month 1 of their lifecycle. Customers in the December 2010 cohort had a Month-1 retention rate of 36.6% while those in the November 2011 cohort had a percentage of 11.2%. Therefore, the data supports the conclusion that the first 30-days from the date of purchase are the periods with the greatest risk for customer churn.

- What to do:

Automate 3 email campaigns with product usage tips, recommendations for complementary products, second-purchase discounts and links to reorder on Days 7, 14, and 28 after the date of first purchase.

- Evidence of New Cohort Behaviour

From the Dec 2010 cohort, Month-1 retention declined (36.6%) to the Month-1 retention of the Nov 2011 cohort (11.2 %.) This indicates that newer cohorts do not tend to buy second purchases.

- Team & Ownership

Product Team and Marketing Team for email automation setup and content and offers.

- Timeline

4 weeks

- Success Metric

To increase the Month-1 retention rate from 11.2% to 20-25% (within 3 cohorts)

- Risk

Low Engagement/Interaction with emails. All subject lines will be A/B tested and all email contents will be personalized.

- Projected Outcomes

Increased customer's repeat buying behaviour, increased customer's retention rate, and increased revenue growth.

Recommendation 2: Prioritize marketing budget to products and countries that offer high retention.

Description of Evidence:

Product analysis (Figure 9) indicates that revenue is highly skewed towards a few products and most of these product with high revenues have low repeat buyer rate (less than 20%). This equates to no returns for these products, and directly affects the repeat rates of purchase found in the cohort analysis. The repeat customer rate is consistently higher in international markets, including the Netherlands, Germany and EIRE, whereas the United Kingdom accounts for more than 80% of the total revenue (Figure 10). The customers from these markets are more loyal after their first purchase. Marketing spending is on volume and not loyalty, which is a waste of marketing budget.

- What to do:

Boost marketing campaign investment in top three EU markets (Netherlands, Germany, and EIRE) by 20%. Highlight products with a high repeat rate, like PICNIC BASKET WICKER 60 PIECES, in the main email and on the home page. Stop or slow the investment in campaigns where customers have been shown to have a < 15% month-1 retention rate.

- Evidence:

Product analysis (Figure 9) shows that the products with the highest revenue all have below 20% repeat buyer rates: PAPER CRAFT LITTLE BIRDIE (£163,000) with a 13% repeat buyer rate, REGENCY CAKESTAND 3 TIER (£143,000) with a 17% repeat buyer rate and WHITE HANGING HEART T-LIGHT HOLDER (£100,000) with a 18% repeat buyer rate. The results of the country analysis (Figure 10) indicate that Netherlands, Germany and EIRE are quicker to return for another purchase than the average UK customer. Spending more money on these high-loyalty portions will yield higher long term revenue per pound.

- Team & Ownership:

Marketing Team -tasked with reallocating marketing campaign budgets, creating briefs for international campaigns, and product feature placement decisions.

- Timeline:

6 weeks to re-allocate budgets, update creative team and deploy new campaigns in key markets.

- Success Metric:

Grow repeat customer by 5% in Netherlands, Germany and EIRE within 90 days of campaign. Improve the proportion of repeat revenue from high retention products by 10-15% over the same time frame.

- Risk:

In the short term a UK campaign spend reduction may result in reduced new customer acquisition volume.

- Mitigation:

The reallocation of the marketing budget will only take place for the part of it dedicated to retention. Acquisition campaigns in the Core UK segment will continue to be fully funded and will not be impacted.

- Projected Outcomes:

International repeat revenue growth will generate a significant revenue diversification, bringing BrightCart Cloud away from being a purely UK based business. Including products with high repeatability will help drive high repeat rates and directly contribute to the Month-1 retention goals in Recommendation 1.

Recommendation 3: win-back at risk high value customers immediately.

Evidence

According to RFM customer segmentation performed on the BrightCart customers, as depicted in Figures 7 and Table 2, there are two types of risky high-value customers: 114 Lost - High Value customers (£223,575 revenue) and 608 About to Sleep customers (£336,037 revenue). In total, these segments have accounted for £559,612 revenue at risk.

These customers used to be loyal to BrightCart Cloud before; hence, their probability of buying from it is higher than that of attracting entirely new customers.

- What to do:

Conduct a personal win-back campaign among these 722 customers. E-mail 1 (Day 1): "We miss you" email with an offer of free shipping. E-mail 2 (Day 7): Offer a 15% discount to customers who do not open their emails. E-mail 3 (Day 14): Give early access to new products to those customers who do not buy.

- Evidence:

As per Figures 7 & 8, 114 Lost -High Value customers (£223,575 revenue) and 608 about to Sleep customers (£336,037 revenue) generate £559,612 revenue at risk.

- Owner: CRM / Marketing Team
- Timeline: 2 weeks
- Success Metric: At-risk buyer purchases within 30 days as a percentage of at-risk buyers. Goal is 20% reactivation rate.

- Risk: Discounts may hurt profit margin, use free shipping as first incentive, and only discount for non-responder.
- Estimated Outcome: 20% reactivation rate would bring in approximately £111,924 in additional revenue and increase customer retention.

9. Monitoring Plan

9.1 Key Performance Investigate KPIs

The use of a Power BI dashboard for this undertaking contains live-type data regarding the 5 key performance indicators which are needed to monitor ongoing progress regarding any recommendations made. The table below indicates how to interpret and take action on whether you are meeting your KPIs or if you are below KPI standards:

Table 3 Monitoring KPI table

KPI Metric	Target	Actions If less than Target	Frequency
Month 1- Return To Purchase Rate	>30%	Review Onboarding Email Immediately	Weekly
Return To Purchase Rate %	>70%	Verify that There Is An Active Win-Back Campaign	Weekly
High Value Loss Count	Decreasing	Review How Much Was Spent on Win-Back Offers	Weekly
Repeat Customers Revenue	Increase	Monitor Marketing Channel Results	Monthly
International Revenue (%)	Growing Share	Increase Number of International Campaigns	Monthly
New Customers in Their First Month retention	>30%	Send Out Onboarding Email When Their First Month Has Passed	Monthly

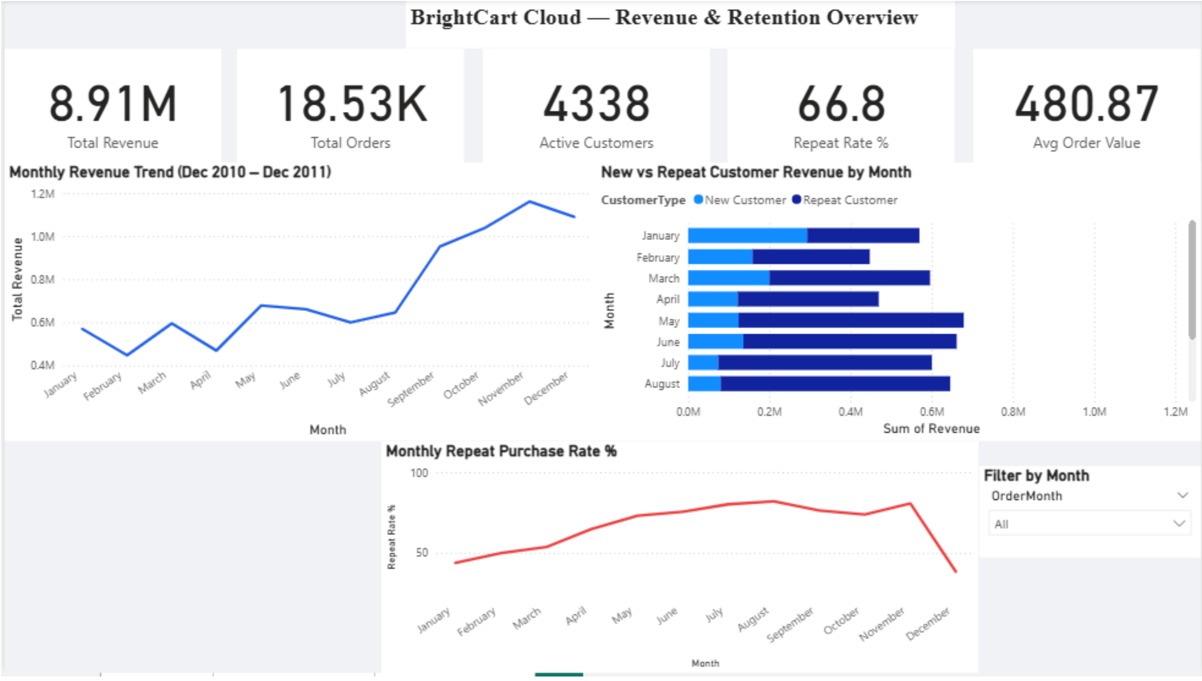

Figure 11 BrightCart Cloud Power BI Dashboard -Overview Page

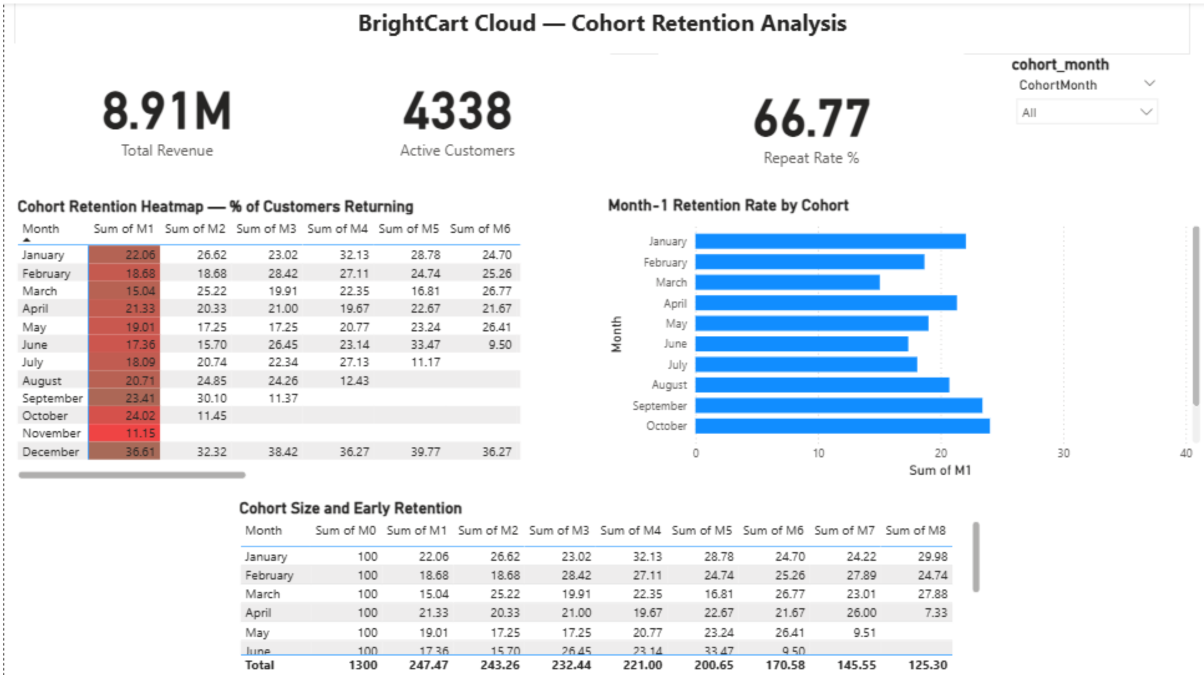

Figure 12 BrightCart Cloud Power BI Dashboard -Cohort Retention Page

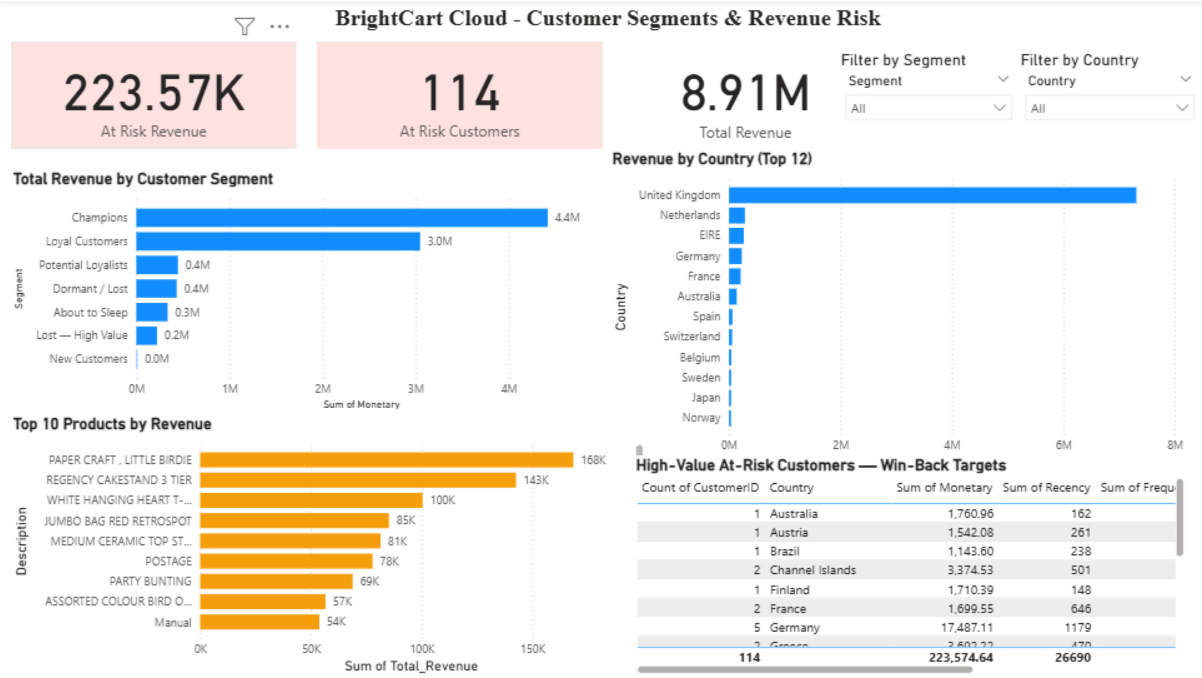

Figure 13 BrightCart Cloud Power BI Dashboard -Customer Segments & Revenue Risk

9.2 Weekly Review Process

Each Monday the Power BI dashboard owner will review the Power BI dashboards and assess the 6 KPIs above. The review process goes as follows:

Step 1: Review Monthly 1-Month Retention Rate. When retention is below 30% Onboarding communications and customer engagement activities are reviewed by the Product and Marketing Teams.

In Step 2, check the Repeat Purchase Rate % column. In Step 2 look at the Repeat Purchase Rate % column. If less than 70% it is a time to review Loyalty, retention and re-engagement campaigns by the Marketing Team.

Step 3: Check Lost -High Value Customer Count. When the number goes up from the current baseline, the CRM Team further increases their win-back efforts and customer personalization.

Step 4: Analyse Trend of Repeat Customer Revenue. When repeat customer revenue does not increase or even decreases, the allocation of a marketing budget and promotional activities are revaluated.

Step 5: Check out the International Revenue Share. In the event of below-predicted international sales growth, further international marketing campaigns are undertaken in priority countries.

Step 6: Review At-Risk Revenue. Targeted recovery campaigns go to dormant and high-value customers if at-risk revenue increases.

All findings are documented on a common KPI tracking sheet and reported on monthly basis. Every Monday, the Marketing Manager, CRM Lead and Product Manager have a 15 minute review meeting to review dashboard performance and campaign progress. An extended 30-minute review meeting is held the first Monday of every month to discuss the results of recommendations, approve corrective action plans and settle budget changes for the upcoming month.